

Student Performance Analysis

Beginner portfolio project

Relate study time, background factors, and scores to identify performance drivers.

project_id: student_performance | template_id: root_cause | source_rows: 48 | prepared_rows: 48 | generated_at: 2026-08-11T06:45:00.266448+00:00

Executive summary

Student Performance Analysis analyzed 48 rows across 7 fields. The reusable root_cause BI template presents source-backed KPIs, drivers, and detail tables. Total study hours is 328.93 with a transparent synthetic validation fixture or uploaded source.

Headline metrics

Rows analyzed

48

Total study hours

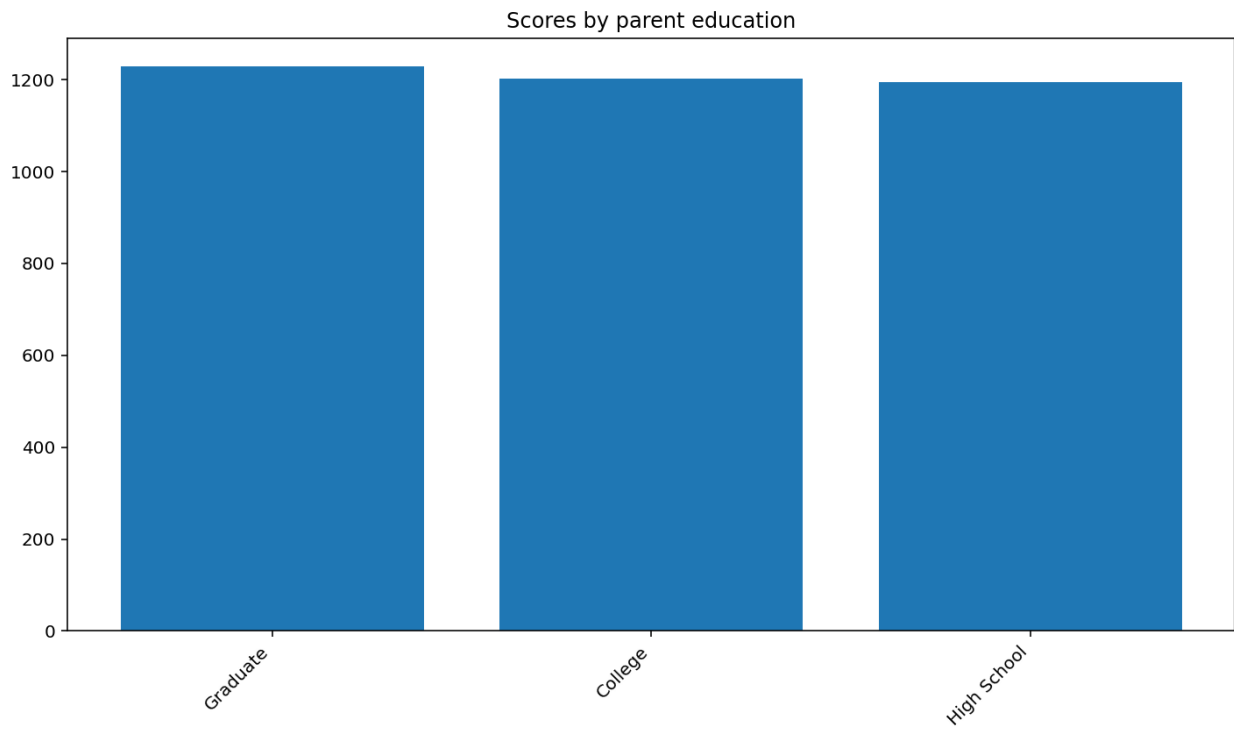
328.93

Average study hours

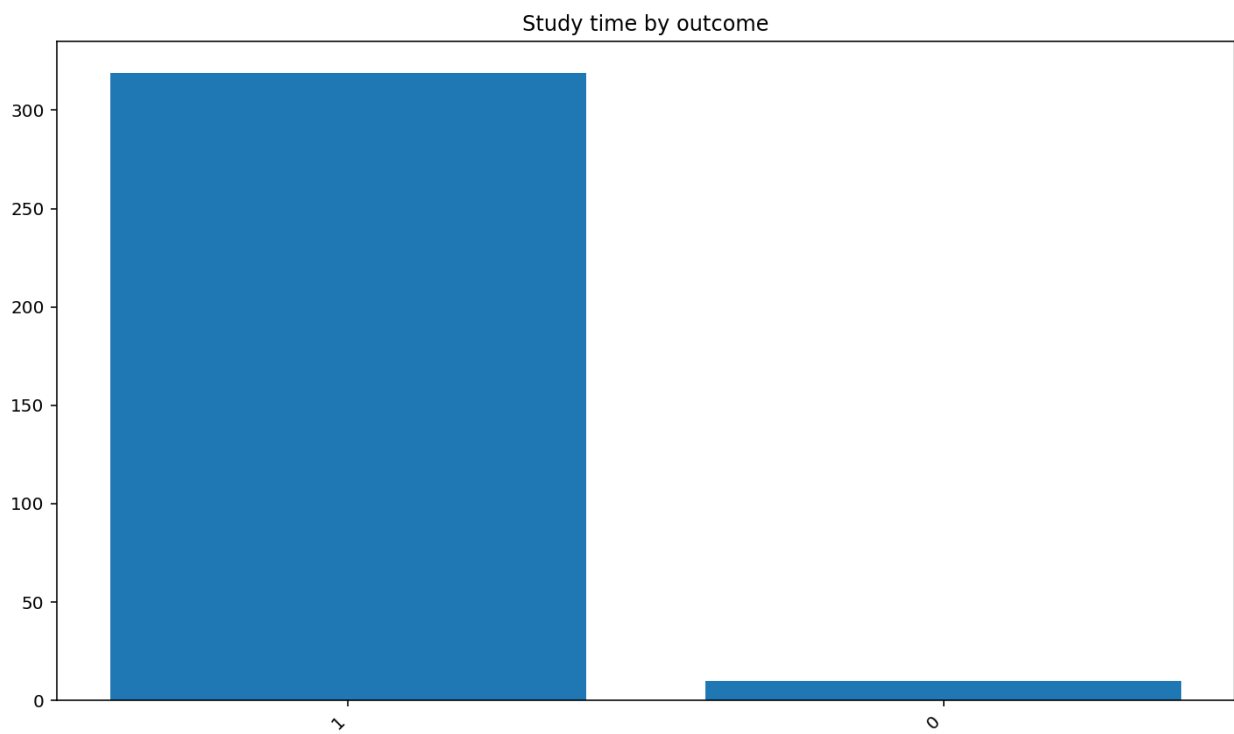
6.85

Drivers and trends

Scores by parent education



Study time by outcome



Detail by parent_education

parent_education	count	sum	mean
Graduate	16	1229.0	76.8125
College	16	1203.0	75.1875

parent_education	count	sum	mean
High School	16	1195.0	74.6875

Detail by passed

passed	count	sum	mean
1.0	43	319.15999999999997	7.422325581395348
0.0	5	9.77	1.954

Key findings

math_score and writing_score are strongly correlated ($r=0.939$).; study_hours and math_score are strongly correlated ($r=0.913$).; study_hours and writing_score are strongly correlated ($r=0.885$).; math_score and reading_score are strongly correlated ($r=0.877$).; reading_score and writing_score are strongly correlated ($r=0.831$).; study_hours and reading_score are strongly correlated ($r=0.807$).; student_id is nearly unique per row (100.0%).; study_hours is nearly unique per row (100.0%).

Caveats and assumptions

Fixtures are synthetic validation data. Uploaded datasets remain the source of truth; field aliases are recorded in the build metadata.